

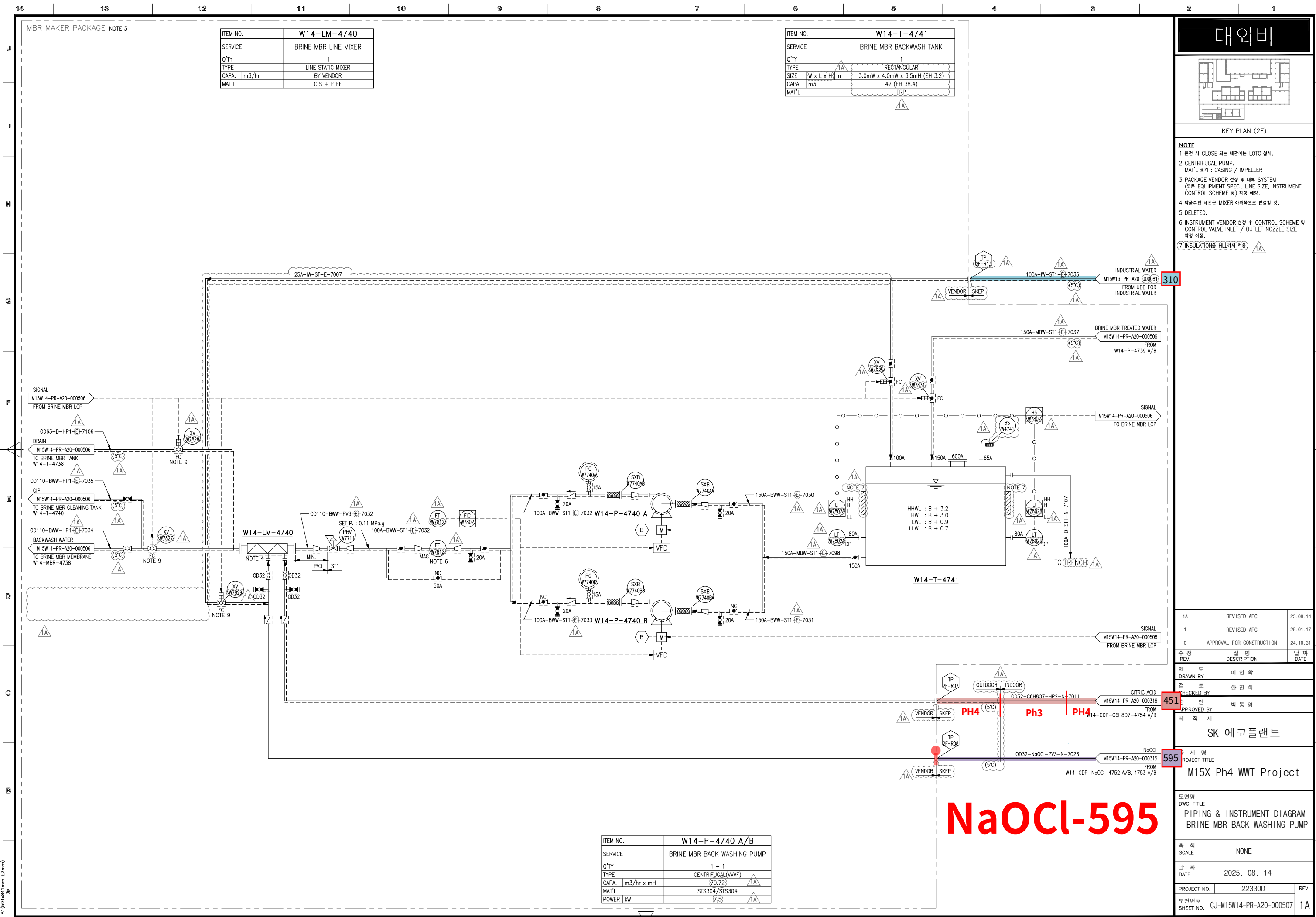
DRAWING LIST FOR TEST PACKAGE

NaOCL-595

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	595	3FA	OD32-NaOCl-PV3-N-7026-01	1 OF 1	0	CJ-M15W14-PR-A20-000507		
2	595	4FA	OD25-NaOCl-PV3-N-7022-01	1 OF 3	0	CJ-M15W14-PR-A20-000315		
3	595	4FA	OD25-NaOCl-PV3-N-7023-01	2 OF 3	0	CJ-M15W14-PR-A20-000315		
4	595	4FA	OD32-NaOCl-PV3-N-7024-01	1 OF 1	0	CJ-M15W14-PR-A20-000315		
5	595	4FA	OD32-NaOCl-PV3-N-7025-01	1 OF 1	0	CJ-M15W14-PR-A20-000315		
6	595	4FA	OD32-NaOCl-PV3-N-7026-01	1 OF 2	0	CJ-M15W14-PR-A20-000315		
7	595	4FA	OD32-NaOCl-PV3-N-7026-02	2 OF 2	0	CJ-M15W14-PR-A20-000507		
8	595	2FA	OD32-NaOCl-PV3-E-7026-01	1 OF 1	1	CJ-M15W14-PR-A20-000507		
9	595	R2FA	OD32-NaOCl-PV3-N-7026-01	1 OF 1	1	CJ-M15W14-PR-A20-000507		
10	595	4FA	OD32-NaOCl-PV3-N-7028-01	1 OF 1	0	CJ-M15W14-PR-A20-000315		



ITEM NO.	W14-P-4740 A/B
SERVICE	BRINE MBR BACK WASHING PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL(WWF)
CAPA.	m3/hr x mH
MAT'L	STS304/STS304
POWER	kw

ITEM NO.	W14-LM-4740
SERVICE	BRINE MBR LINE MIXER
Q'TY	1
TYPE	LINE STATIC MIXER
CAPA.	m3/hr
MAT'L	C.S + PTFE

ITEM NO.	W14-T-4741
SERVICE	BRINE MBR BACKWASH TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	3.0mW x 4.0mW x 3.5mH (EH 3.2)
CAPA.	42 (EH 38.4)
MAT'L	FRP

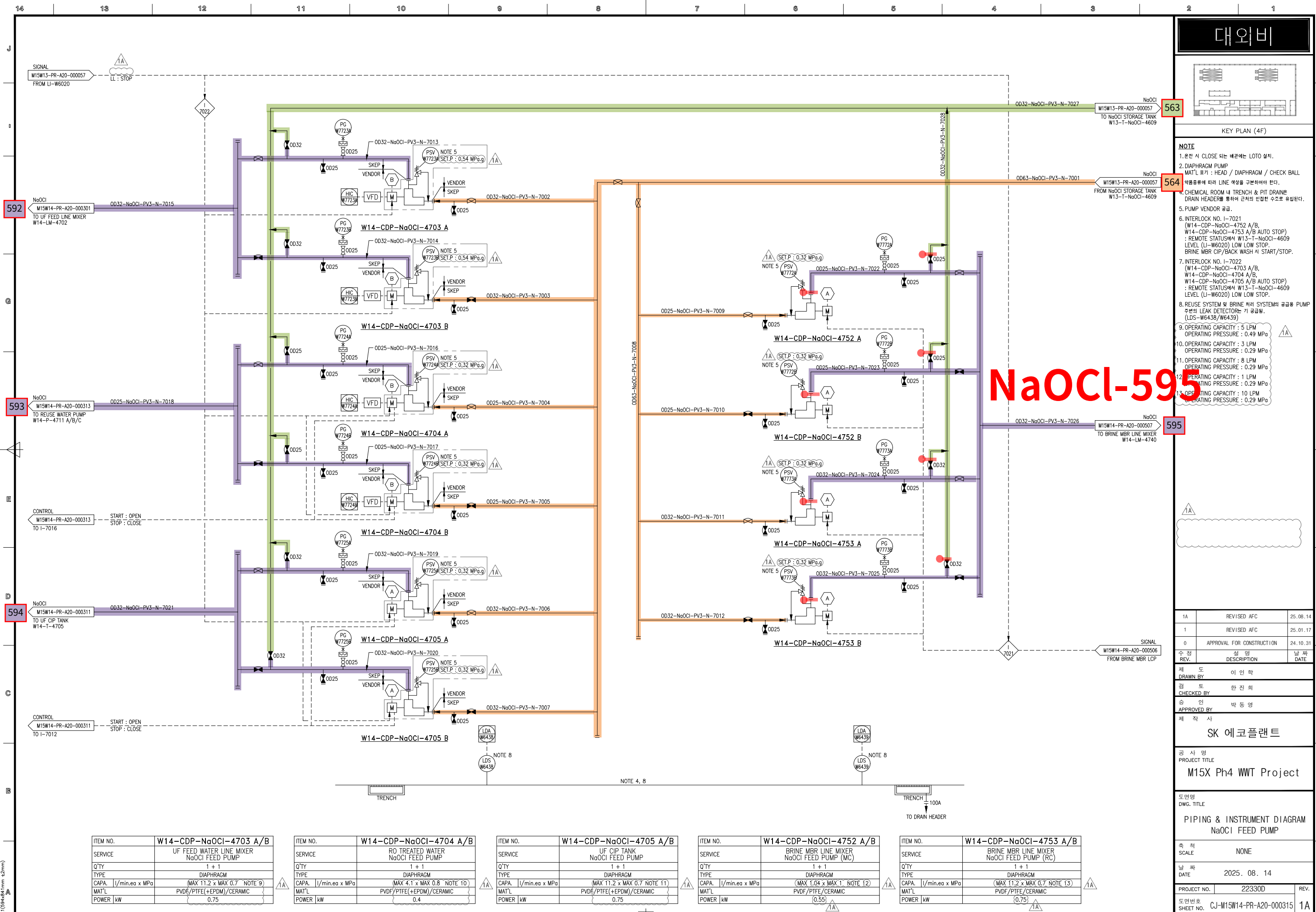
대외비

KEY PLAN (2F)

NOTE

1. 운전 시 CLOSE 되는 배관에는 LOTO 설치.
2. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
3. PACKAGE VENDOR 선정 후 내부 SYSTEM
(모든 EQUIPMENT SPEC., LINE SIZE, INSTRUMENT
CONTROL SCHEME 등) 확정 예정.
4. 약품주입 배관은 MIXER 아래쪽으로 연결할 것.
5. DELETED.
6. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및
CONTROL VALVE INLET / OUTLET NOZZLE SIZE
확정 예정.
7. INSULATION을 HLL까지 적용

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.10.31
수정 REV.	실명 DESCRIPTION	날짜 DATE
제 DRAWN BY	이인학	
검 TO	한진희	
인 CHECKED BY	박동영	
사 APPROVED BY		
제 작 사	SK 에코플랜트	
사 명	M15X Ph4 WWT Project	
도면명 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM BRINE MBR BACK WASHING PUMP	
속 척	NONE	
날 DATE	2025. 08. 14	
PROJECT NO.	22330D	REV.
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000507	1A



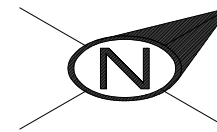
대외비

KEY PLAN (4F)

NOTE

1. 본선 시 CLOSE 되는 배관에는 LOTO 설치.
2. DIAPHRAGM PUMP
MAT'L 표기 : HEAD / DIAPHRAGM / CHECK BALL
작동종류에 따라 LINE 역상을 구분하여야 한다.
3. CHEMICAL ROOM 내 TRENCH & PIT DRAIN
DRAIN HEADER를 통하여 근처의 인접한 수조로 유입된다.
4. PUMP VENDOR 공급.
5. INTERLOCK NO. I-7021
(W14-CDP-NaOCl-4752 A/B,
W14-CDP-NaOCl-4753 A/B AUTO STOP)
: REMOTE STATUS에서 W13-T-NaOCl-4609
LEVEL (LI-W6020) LOW LOW STOP.
BRINE MBR CIP/BACK WASH 시 START/STOP.
6. INTERLOCK NO. I-7022
(W14-CDP-NaOCl-4703 A/B,
W14-CDP-NaOCl-4704 A/B,
W14-CDP-NaOCl-4705 A/B AUTO STOP)
: REMOTE STATUS에서 W13-T-NaOCl-4609
LEVEL (LI-W6020) LOW LOW STOP.
7. REUSE SYSTEM 및 BRINE 처리 SYSTEM의 공급용 PUMP
부분의 LEAK DETECTOR는 기 공급함.
(LDS-W6438/W6439)
8. OPERATING CAPACITY : 5 LPM
OPERATING PRESSURE : 0.49 MPa
9. OPERATING CAPACITY : 3 LPM
OPERATING PRESSURE : 0.29 MPa
10. OPERATING CAPACITY : 8 LPM
OPERATING PRESSURE : 0.29 MPa
11. OPERATING CAPACITY : 1 LPM
OPERATING PRESSURE : 0.29 MPa
12. OPERATING CAPACITY : 10 LPM
OPERATING PRESSURE : 0.29 MPa

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.10.31
수정	실명	날짜
REV.	DESCRIPTION	DATE
제 도	이 인 학	
검 토	한 진 희	
승 인	박 동 영	
APPROVED BY		
제 작 사		
SK 에코플랜트		
공 사 명	PROJECT TITLE	
M15X Ph4 WWT Project		
도면명	DWG. TITLE	
PIPING & INSTRUMENT DIAGRAM NaOCl FEED PUMP		
속 측	SCALE	
NONE		
날 짜	DATE	
2025. 08. 14		
PROJECT NO.	22330D	REV.
도면번호	SHEET NO. CJ-M15W14-PR-A20-000315 1A	

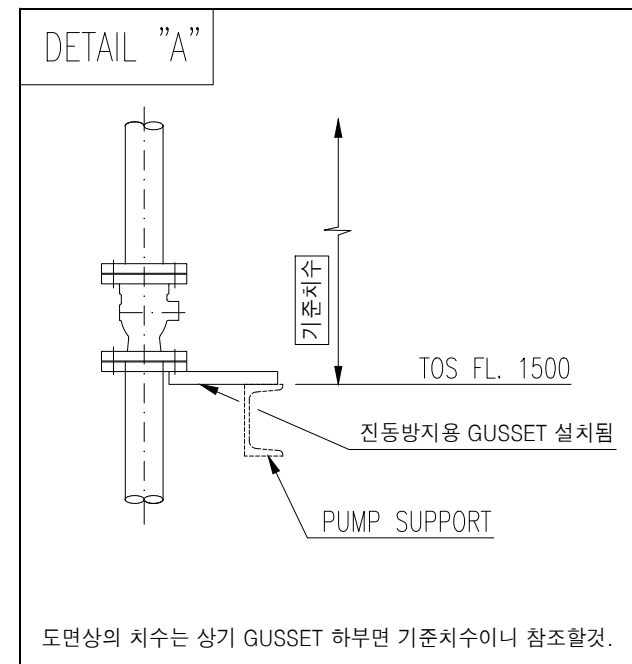
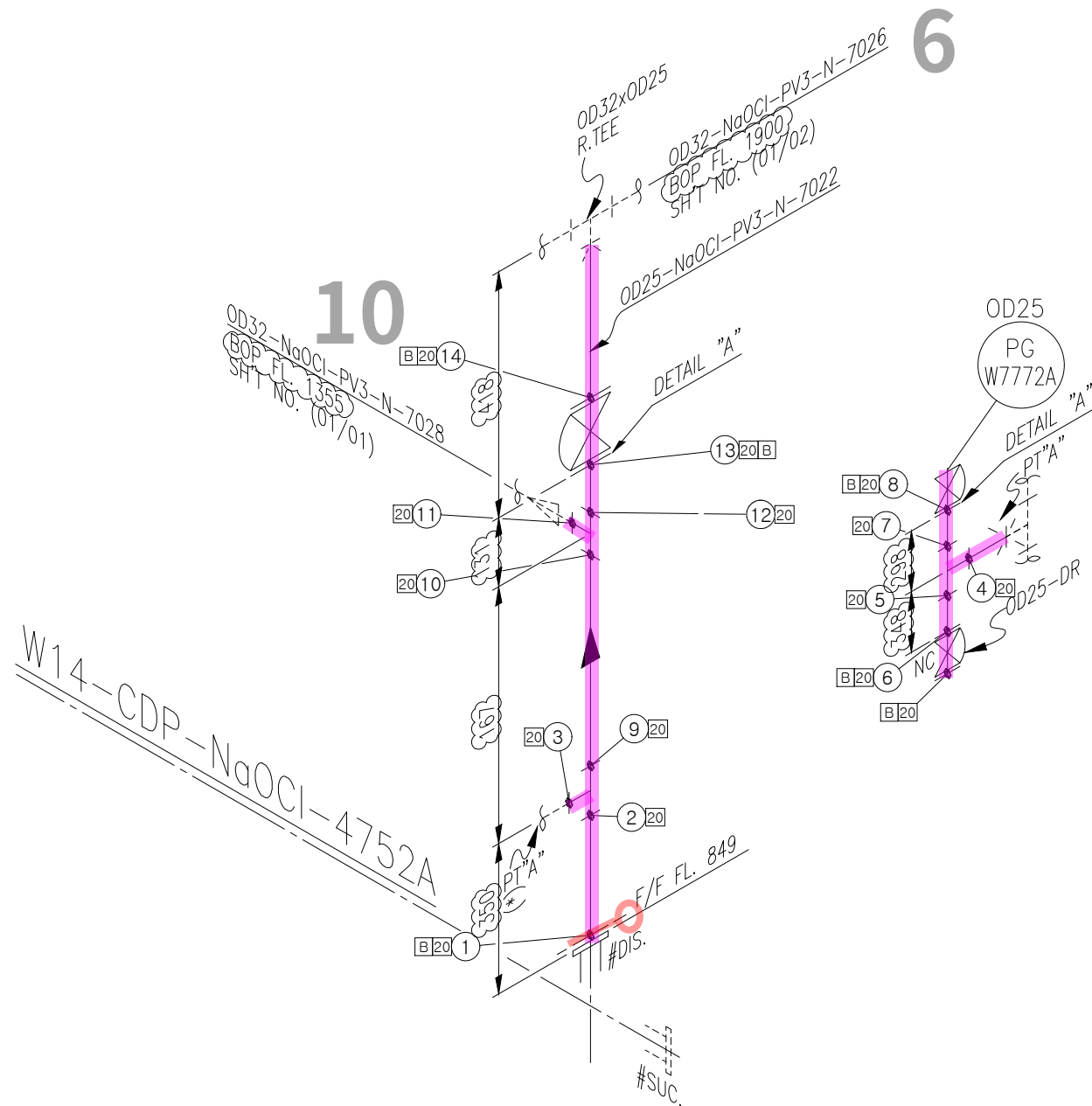


CLIENT  1

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



ITEM	SIZE	Q'TY	DESCRIPTION
PV3			
P	OD32	1.3M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
T	OD25	3EA	TEE EQUAL, CPVC, PE, SOCKET FUSION, SDR13.5
F	OD25	5EA	FLG, KS 10K, FF, STUB FLG CPVC /
			BACKING RING(PP)+INSERT(STEEL), SDR13.5
FB	OD25	1EA	FLG BLIND, KS 10K, FF, CPVC, SDR13.5
G	OD25	6EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B	OD25	6SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
VD	OD25	3EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-CPVC /
			TRIM-PDDM, BB, HW
SH	OD25	6EA	SHIELD
LOTO	OD25	1EA	LOTO
PG	OD25	1EA	PRESSURE GAUGE, KS 10K, FF



<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB10/VB2/VB210	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD10	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						CONTRACTOR 	PROJECT NAME		
	THICK.	-	DESIGN	3.3	60	UNIT 4FA							M15X Ph4 Project		
			OPERATING	3.0	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040010-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY NaOCl	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000315							REV. NO.	DATE	DESCRIPTION

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



LINE NO.	0D32-NaOCl-PV3-N
----------	------------------

4

[illegible]



LINE NO.	0D32-NaOCl-PV3-N
----------	------------------

5

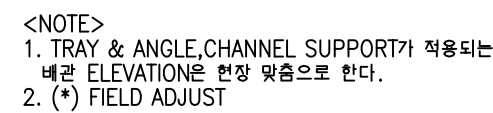


CLIENT		 6	
CONTRACTOR 		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY NaOCI		JOB NO.	
		22330D	
LINE NO. 0D32-NaOCI-PV3-N-7026		SH'T NO.	REV.
		01 / 03	0



CLIENT  9

	CONTRACTOR		PROJECT NAME	
			M15X Ph4 Project	
00	COMMODITY	NaOCI	JOB NO.	
00			22330D	
D	LINE NO.	0D32-NaOCI-PV3-E-7026	SH'T NO. 01 / 01	REV. 1

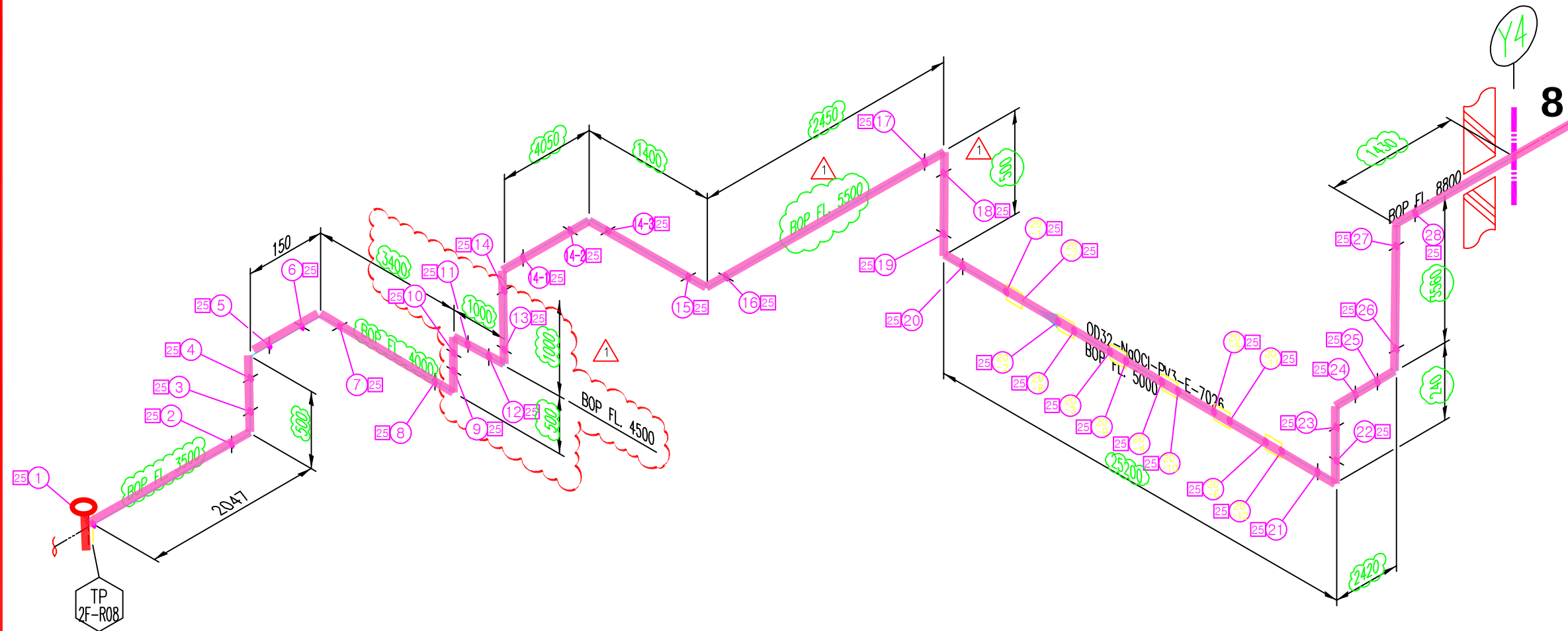


* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB10/VB1/VB10/VB2/VB20	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

[illegible]



ITEM	SIZE	Q'TY	DESCRIPTION
PV3			
P	OD32	49.4M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
E90	OD32	15EA	ELBOW 90 DEG, CPVC, PE, SOCKET FUSION, SDR13.5
F	OD32	1EA	FLG, KS 10K, FF, STUB FLG CPVC /
			BACKING RING(PP)+INSERT(STEEL), SDR13.5
INC	32xOD32	49.4M	관 보존재(고무보링) + 알루미늄 CASING



<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

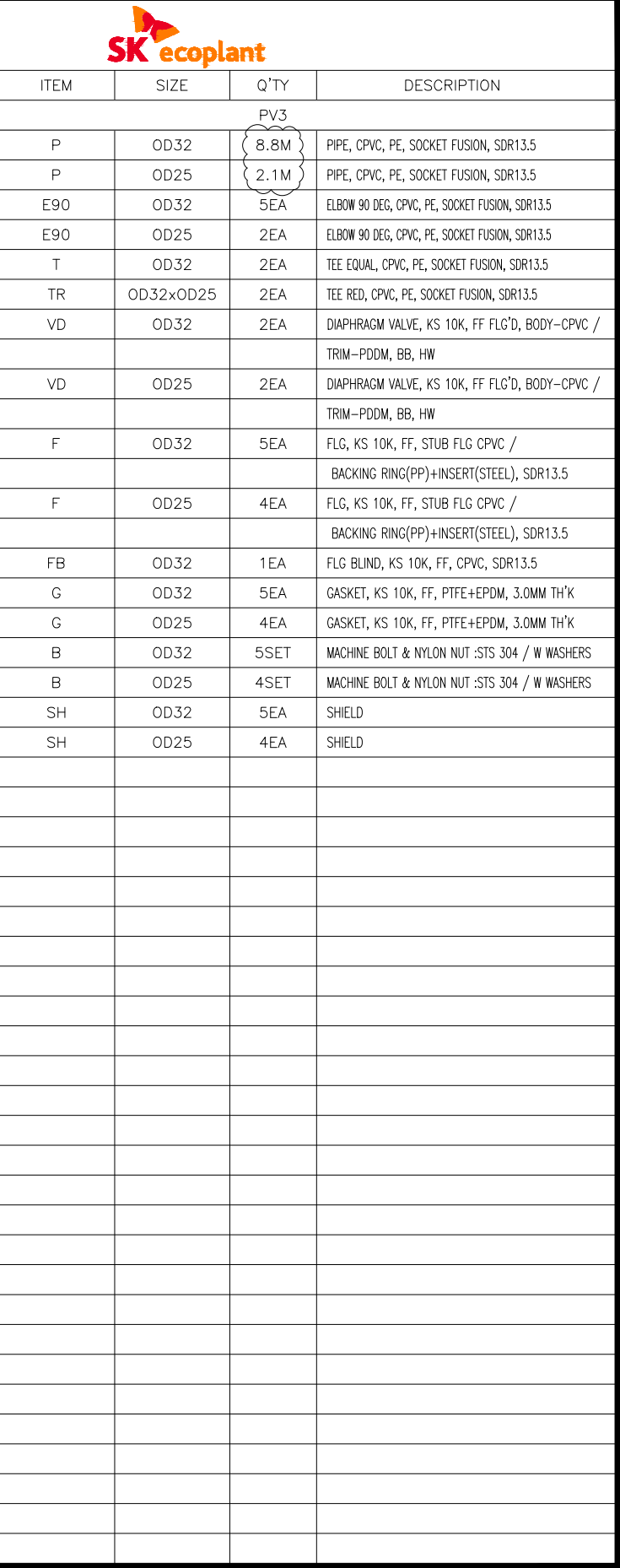
* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB10/VB2/VB2L0	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

CLIENT



9

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME	
	THICK.	25	DESIGN	10	60	UNIT R2FA								M15X Ph4 Project	
			OPERATING	7.5	AMB	REF. PLAN DWG.		25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-020016-C		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	NaOCl	22330D	
						REF. P & ID	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
					CJ-M15W14-PR-A20-000315,000507								OD32-NaOCl-PV3-E-7026	01 / 01	1



CLIENT		10	
			
CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
NaOCl		22330D	
LINE NO.		SH'T NO.	REV.
0D32-NaOCl-PV3-N-7028		01 / 01	0